Executive Summary

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Five Year Survival Prediction of The Breast Cancer Patients from UK and Canada Based on the Clinical Information

Project members: Gopal Narayan Srivastava, Monika Pandey, Mehri Mehrnia

GitHub repository: GitHub

Problem Description

Breast cancer is the most common cancer diagnosed in women [1], and doctors lean heavily on the "5-year survival" mark when they talk to patients about what to expect, decide how often to see them for follow-up, and choose how aggressive treatment should be [2]. But 5-year outcomes are far from uniform as some patients with aggressive tumors end up surviving for longer than 5 years, on the other hand, others with non-agressive sometimes show frequent relapse. Thus, figuring out patients outcome can help doctors making treatment decisions, inform patients of their options, and insurance companies in making policies and financial decisions related to patients with breast cancers. For our study the stakeholders include patients, hospitals, pharma and insurance companies.

Project Aims

To explore into this, we asked two specific questions.

1. Will the patient survive at least 5 years after being diagnosed?
2. Is there any difference in survival of TNBC (Basal) cancer subtype compared to other subtypes?

Note: We define "surviving 5 years" as being alive 60 months or more after diagnosis.

Dataset

Clinical data of 2,509 female breast-cancer patients from the UK and Canada, was downloaded from cBioPortal [3]. From the dataset we removed 528 patients as their survival status along with 10 clinical features were missing.

Feature Selection, Model Selection, and Performance

Since, we framed our task as a binary classification problem focused 5 years survival (positive class: "survived $\geq$ 5y"), we focused on patients from the METABRIC [3] cohort whose 5-year outcome was unambiguously known. Patients with living survival status and overall survival less than 5-year mark were excluded, since in medicine 5-year survival is the gold standard. Our final analytical dataset included 1,917 patients drawn from 5 cohorts, of whom roughly 77% survived 5 years and 33% died within 5 years.

1. In our study, we made use of the fact that METABRIC's patients come from 5 distinct cohorts and split by cohort: cohorts 1–3 (1,518 patients) formed the development pool and cohorts 4 and 5 (399 patients) were separated as an external holdout.
2. Missing values in the clinical fields were filled using Multivariate Imputation by Chained Equations (MICE), per cohort so that no information leaked across the cohort boundary before splitting.

For feature selection, we started with the imputed data then ranked candidate features by importance in a preliminary *RandomForest* screen and performed elbow analysis to get top 11 features: *Age at Diagnosis, Nottingham prognostic index, hetloss, gain, Tumor Size, amp, TMB (nonsynonymous), homdel, Integrative Cluster, Pam50 + Claudin-low subtype, and Type of Breast Surgery.*

For model selection, we used leave-one-cohort-out cross-validation on the development pool (3-fold GroupKFold, 3 Cohorts in Dev pool) to compare four classifiers: (a) Logistic Regression with L1/L2 penalties, (b) LightGBM, (c) XGBoost, and (d) Support Vector Machine with an RBF kernel. Class imbalance was handled inside each pipeline as follows: *class_weight='balanced'* for LR and SVM, *scale_pos_weight = n_neg / n_pos* for LightGBM and XGBoost, so that the minority negative ("died <5y") class contributed equally to the loss. Each model was tuned by GridSearchCV scored on ROC-AUC and refit on the entire dev pool with its best hyperparameters before ever touching the holdout.

The four tuned models produced very similar cross-validated ROC-AUCs on the dev pool, with an ∼0.03 spread between the strongest and weakest. On the sealed holdout, we then reported ROC-AUC, PR-AUC, MCC, and per-class precision, recall, and F1:

Model	ROC-AUC	PR-AUC	MCC	Precision (died / survived)	Recall (died / survived)	F1 (died / survived)
Logistic Regression	0.76	0.89	0.36	0.40 / 0.91	0.80 / 0.63	0.53 / 0.74
LightGBM	0.76	0.91	0.39	0.43 / 0.91	0.77 / 0.69	0.55 / 0.78
XGBoost	0.77	0.91	**0.40**	0.43 / 0.91	0.78 / 0.69	**0.56** /0.78
SVM (RBF kernel)	**0.79**	**0.92**	0.38	0.42 / 0.91	0.79 / 0.66	0.54 / 0.76

Table 1: Model Evaluation Results on Unseen Cohorts 4, 5

Key Takeaways

1. The RBF-SVM achieved the best threshold-independent ranking (ROC-AUC 0.79, PR-AUC 0.92), meaning it performs best in ordering patients from lowest to highest risk of dying within 5 years.
2. XGBoost achieved the best balanced classification score (MCC 0.40) and F1 on the clinically important minority class, meaning it makes the best individual decisions (yes/no calls) at 0.5 probability threshold.
3. Since, the gap in performance between linear and non-linear models is small (≈0.03 AUC), it signals that most of the predictive information in these features is linear.
4. Feature importance analysis identified age at diagnosis, the Nottingham Prognostic Index, tumor size, and the Pam50 + Claudin-low molecular subtype as the strongest predictors. Copy-number Alterations (CNA), mainly gain and hetloss and the Integrative Cluster labels contributed additional signal beyond what standard clinical features alone would provide.
   (a) Standard clinical prognostic factors still carry the most weight. Age, tumor size, grade, and NPI, dominate the model, confirming that decades-old clinical intuition is not far off from what a modern ML model would use.
   (b) Molecular subtype adds meaningful information beyond staging. Pam50 + Claudin-low classification contributed independent signal to the prediction, consistent with the growing role of subtype-directed therapy in the clinic.
   (c) Copy-number burden matters at the genome-wide level. The four CNA were more useful than individual gene-level events, suggesting overall genomic instability is a more robust signal than any single alteration for predicting 5-year outcome.

In conclusion, our model provides an effective means of distinguishing between breast cancer patients who will and will not survive 5 years from diagnosis, using only information available at the time of that diagnosis.

Future work

1. We would replace binary classification with survival analysis such as Cox proportional hazards or DeepSurv to use censored patients rather than drop them.
2. Incorporate gene expression signatures alongside the clinical and copy-number features, validate the model on a non-METABRIC cohort such as TCGA-BRCA to test generalization beyond a single consortium.
3. For clinical use, tune the decision threshold to reflect the relative cost of false negatives versus false positives in downstream care decisions

References

[1] World Health Organization. Breast cancer fact sheet. https://www.who.int/news-room/fact-sheets/detail/breast-cancer, 2024. Accessed: 2026-07-12.

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[3] cBioPortal for Cancer Genomics. Breast cancer (metabric, nature 2012 & nat commun 2016) study summary. https://www.cbioportal.org/study/summary?id=brca_metabric, 2026. Accessed: 2026-07-12.